

Kartik Koul

SOFTWARE DEVELOPMENT ENGINEER (SDE) · COMPUTER APPLICATIONS & SCIENCE (MASTERS)

✉ kartikkoulwork@gmail.com 🌐 kartikkoul.com 🐙 [github/kartikkoul](https://github.com/kartikkoul) [in linkedin/kartikkoul](https://www.linkedin.com/in/kartikkoul)

Skills –

Programming :	Javascript · Typescript · Python · C++
Frontend :	React · Redux Toolkit · Next.js · NodeJS · TailwindCSS · Framer Motion
Backend :	Node.js · Express.js · FastAPI · REST APIs · WebSockets · Async programming · Authentication(JWT/OAuth) · Background Jobs
AI :	RAG · Agentic AI · Prompt Engineering · LLM Applications · Generative AI · Multimodal AI · Semantic Search · Vector Search
AI Frameworks :	LangChain · LangGraph
Databases :	Postgres · MongoDB · Mongoose · Prisma · Pinecone · pgvector
Cloud & DevOps :	Docker · AWS · GCP · Monorepo (Turborepo) · CI/CD · Vercel
Tools :	Bash · VS Code · Cursor · Git · GitHub · npm · Postman · uv · Figma

Experience -

SDE-I , Advanced Voice Research Labs, AVRL

Presently Working

Tech Stack: Python, TypeScript, JavaScript, FastAPI, Flask, Next.js, React, PostgreSQL, Prisma, MongoDB, Mongoose,

REST APIs, WebSockets, LangChain, LangGraph, RAG, Docker, AWS, Git

April 2022 – Present

- Engineered multiple full-stack internal products using Next.js, FastAPI and PostgreSQL, reducing operational bottlenecks across engineering and business teams.
- Architected and developed an AI-powered Employee Task & Progress Management platform with role-based access control, automated workflows and analytics dashboards, significantly improving project visibility.
- Built an enterprise Retrieval-Augmented Generation (RAG) assistant over internal documentation, client knowledge, project artifacts and operational data, enabling engineers to retrieve contextual information within seconds instead of manual searches.
- Developed modular AI workflows using LangChain and LangGraph, enabling multi-step reasoning, tool calling and agent-based execution for internal automation.
- Designed and implemented REST APIs and asynchronous backend services using FastAPI & Express.js, improving scalability and reducing API response latency by approximately 40% for internal applications.
- Integrated third-party enterprise platforms including Monday.com and Zendesk, enabling seamless synchronization of tickets, tasks and operational data that cut the manual work upto 70%.
- Automated complex logistics workflows by building services and bots, streamlining high-volume operational processes.
- Optimized existing automation pipelines, boosted the efficiency to upto 60%, reducing average task execution time from ~60 seconds to ~25 seconds through algorithmic improvements and workflow redesign.
- Designed reusable backend modules and shared libraries adopted across multiple internal applications, accelerating feature development and improving maintainability.
- Refactored legacy backend modules into reusable and maintainable components, reducing production issues by approximately 95%.
- Diagnosed and resolved production incidents based on client's tickets or internal tickets, improving application reliability and minimizing errors, bugs and downtime.

Web Developer , Tech Table (Internship)

7 Months

Tech Stack: Javascript, Python, CSS3, HTML5, React, Redux, Django, Bootstrap, RestAPI

January 2021 – July 2021

- Developed responsive web applications using React, Django and REST APIs for internal and client-facing projects.
- Built reusable UI components and integrated backend APIs, improving development efficiency.
- Delivered features across 3 production projects and 4 internal applications in an Agile environment.
- Delivered technical seminars on UI Optimization, Animations and Frontend Performance for new engineering interns.

Education -

Amity University

Masters in Computer Applications, 8.2 CGPA

Noida, Uttar Pradesh

Jul 2022 – Aug 2024

University of Jammu

Bachelors in Computer Applications and Science, 7.3 CGPA

Jammu, J&K

August 2018 – October 2021

Projects –

My KnowledgeBase

Tech Stack

<https://github.com/kartikkoul/rag-pdf-qna>

Python | FastAPI | NextJS | PostgreSQL | Prisma | Langchain | Docker | AWS | Vercel

My Knowledgebase is a full-stack Retrieval-Augmented Generation (RAG) & AI powered application that lets users upload PDF documents to build a private, personal knowledge base, then ask questions that are answered strictly from the uploaded content with token-by-token streaming, hybrid retrieval, and per-user data isolation.

SketchMates

Tech Stack

<https://github.com/kartikkoul/sketchmates>

NextJS | NodeJS | ExpressJS | WebSockets | PostgreSQL | Prisma | Turborepo | AWS

Sketchmates is a collaborative real-time whiteboard using WebSockets, Fabric.js and PostgreSQL supporting concurrent users, live synchronization and persistent workspaces.